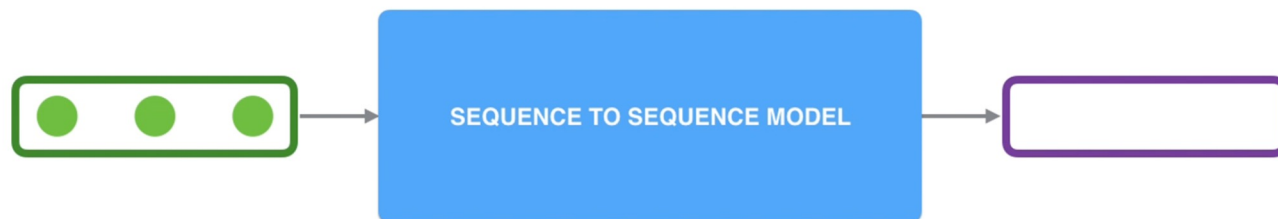


Agenda:

1. Attention Mechanism
2. Transformers
3. Language Models

Sequence-to-Sequence model

A sequence-to-sequence model is a model that takes a sequence of items (words, letters, features of and images...etc) and outputs another sequence of items.



Encoder Decoder model

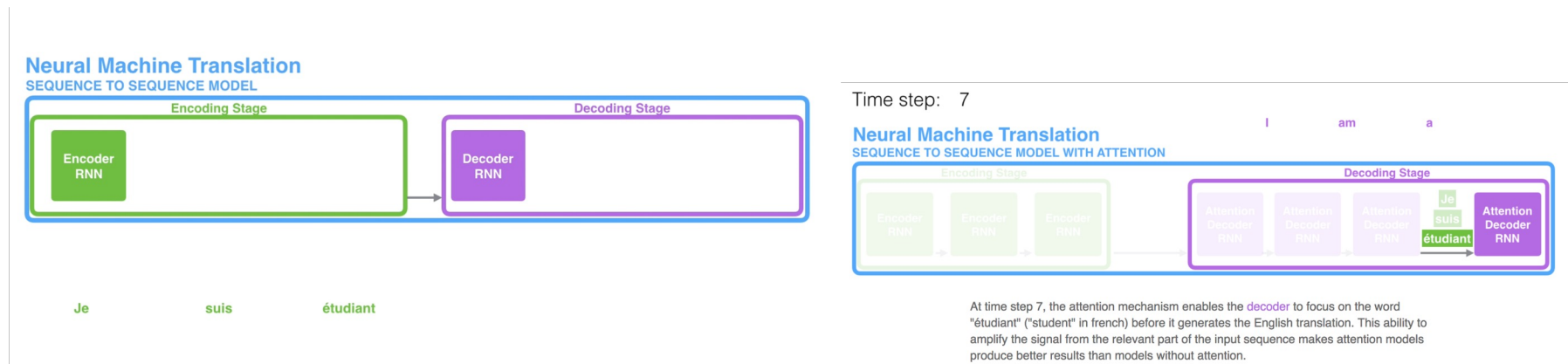
A sequence-to-sequence model is a model that contains an **Encoder** and **Decoder**.

The encoder processes each item in the input sequence, it compiles the information it captures into a vector (called the **context**).

After processing the entire input sequence, the encoder send the context over to the decoder, which begins producing the output sequence item by item.

Attention Please !!!

The context vector turned out to be a bottleneck for these types of models. It made it challenging for the models to deal with long sentences. Attention is for rescue!



Attention Is All You Need

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Ashish Vaswani*
Google Brain
avaswani@google.com

Noam Shazeer*
Google Brain
noam@google.com

Niki Parmar*
Google Research
nikip@google.com

Jakob Uszkoreit*
Google Research
usz@google.com

Llion Jones*
Google Research
llion@google.com

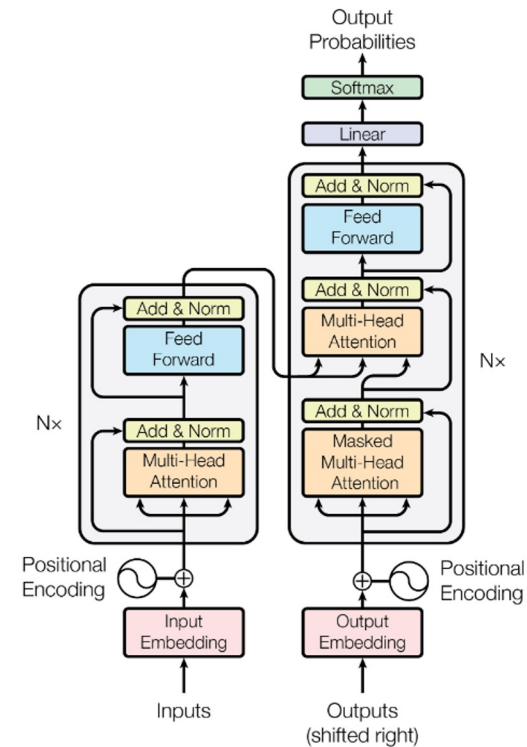
Aidan N. Gomez*[†]
University of Toronto
aidan@cs.toronto.edu

Łukasz Kaiser*
Google Brain
lukasz.kaiser@google.com

Illia Polosukhin*[‡]
illia.polosukhin@gmail.com

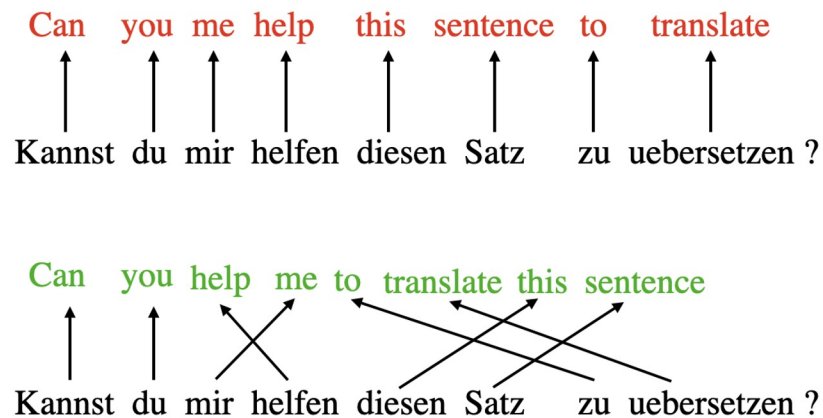
Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.



Self - Attention

The concept of “attention” in deep learning has its roots in the effort to improve Recurrent Neural Networks (RNNs) for handling longer sequences or sentences. For instance, consider translating a sentence from one language to another. Translating a sentence word-by-word does not work effectively.



Attention Mechanism

Attention Visualizations

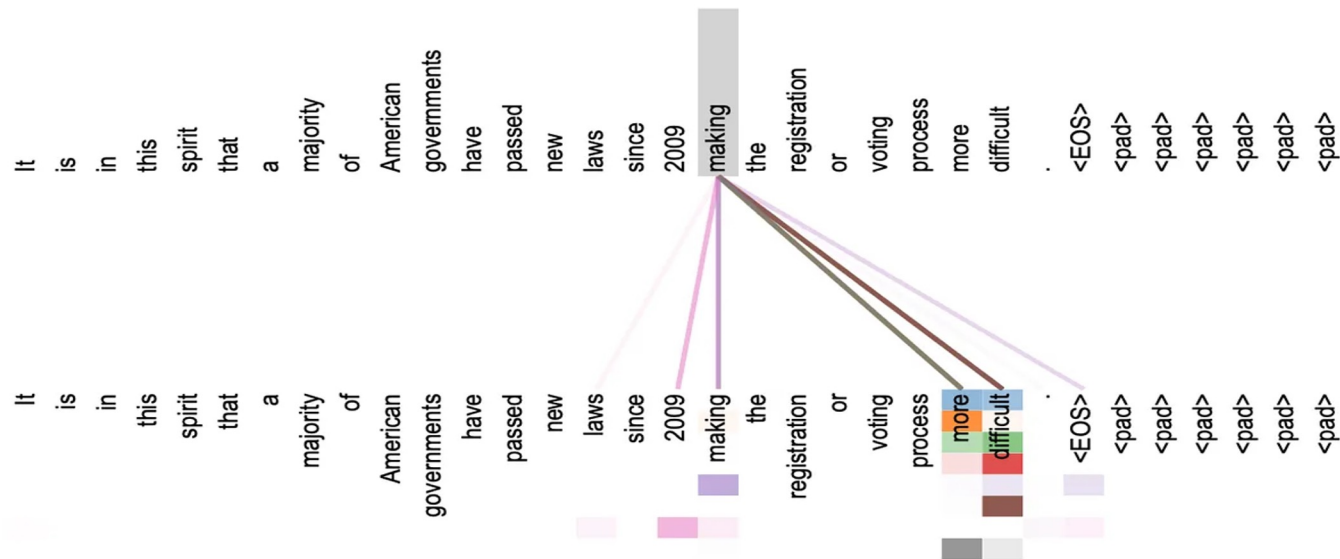
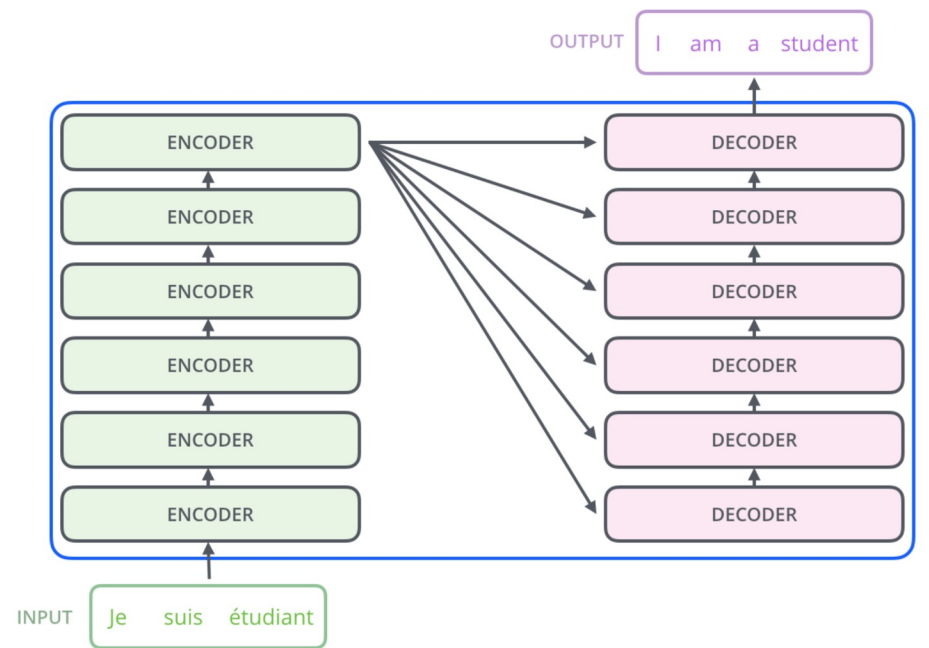
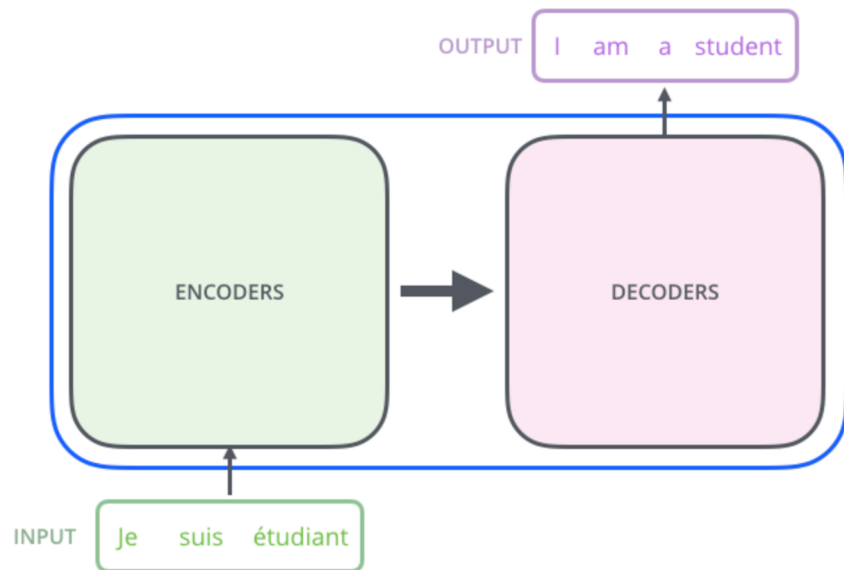
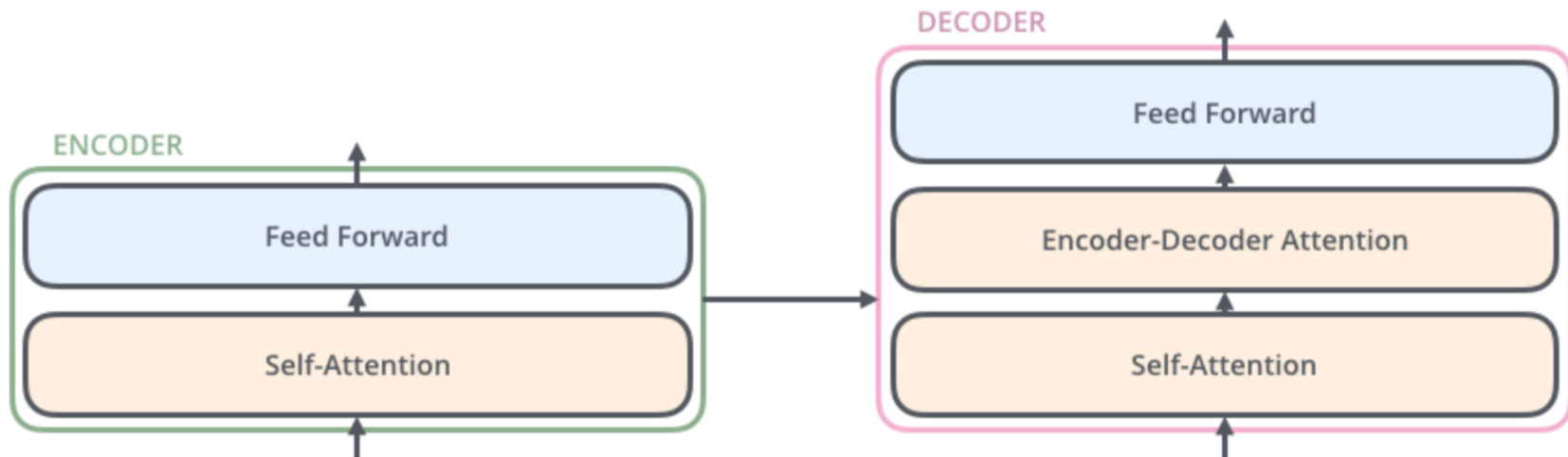


Figure 3: An example of the attention mechanism following long-distance dependencies in the encoder self-attention in layer 5 of 6. Many of the attention heads attend to a distant dependency of the verb 'making', completing the phrase 'making...more difficult'. Attentions here shown only for the word 'making'. Different colors represent different heads. Best viewed in color.

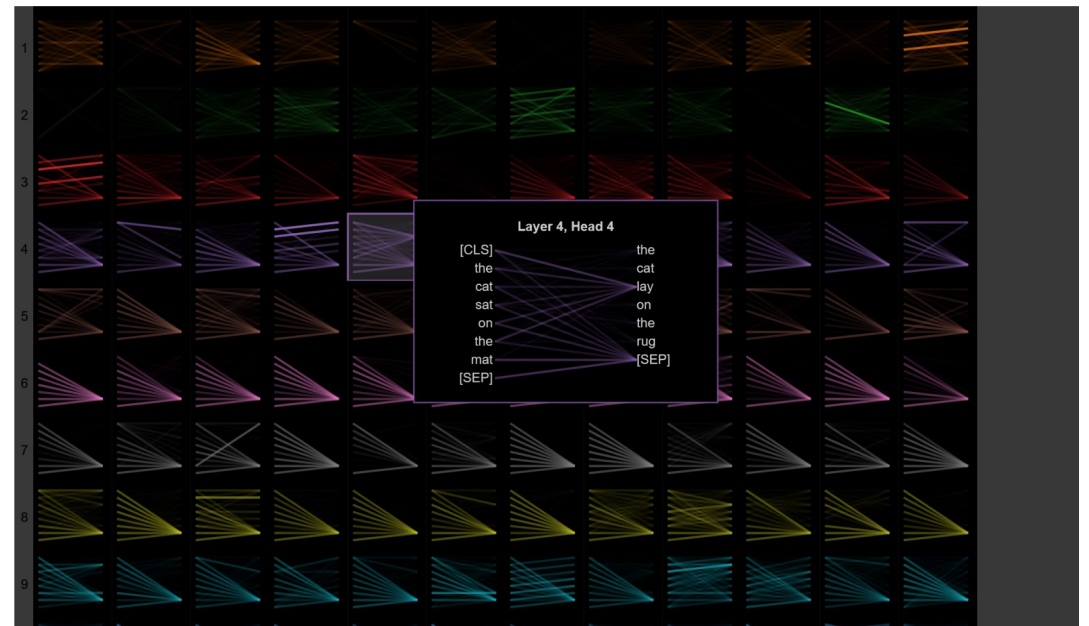
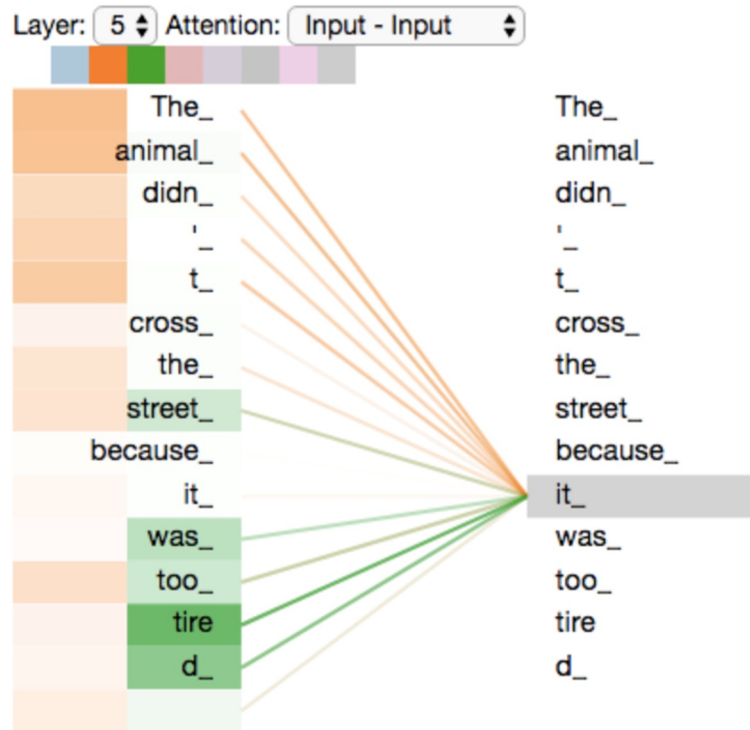
Transformer



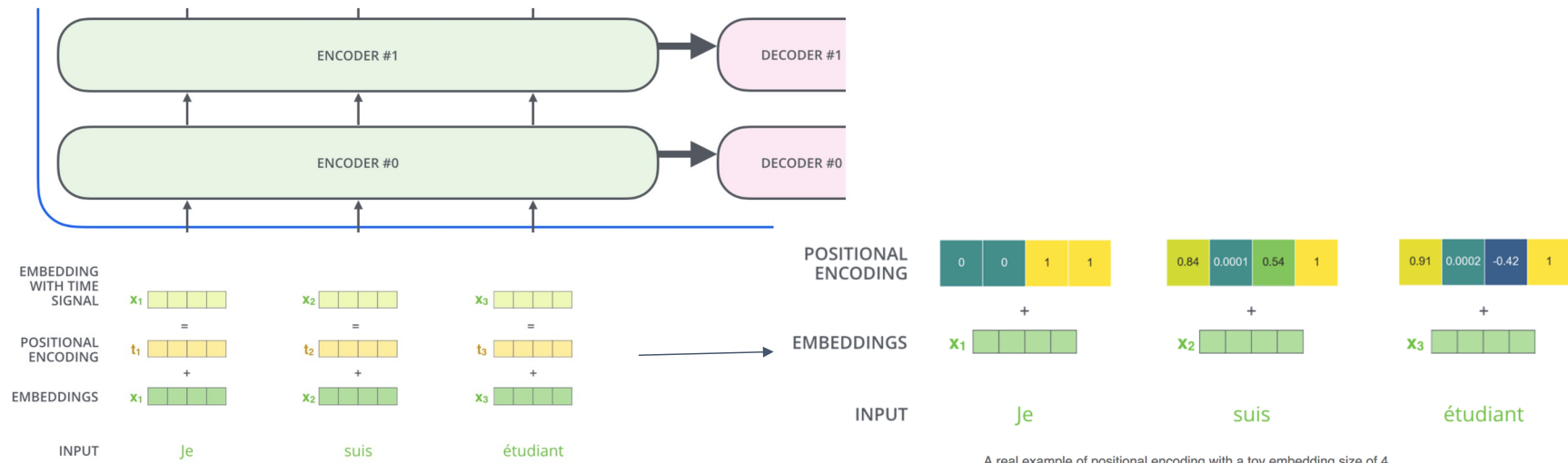
Transformer Encoder-Decoder



Multi-Attention Heads

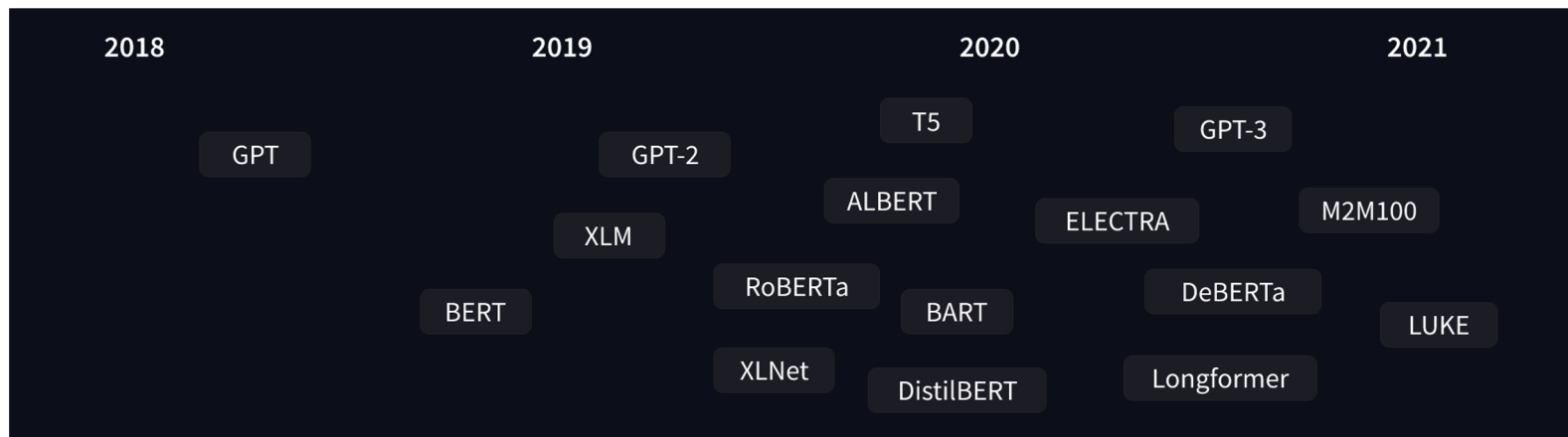


Positional Embeddings



To give the model a sense of the order of the words, we add positional encoding vectors -- the values of which follow a specific pattern.

Transformer Evolution

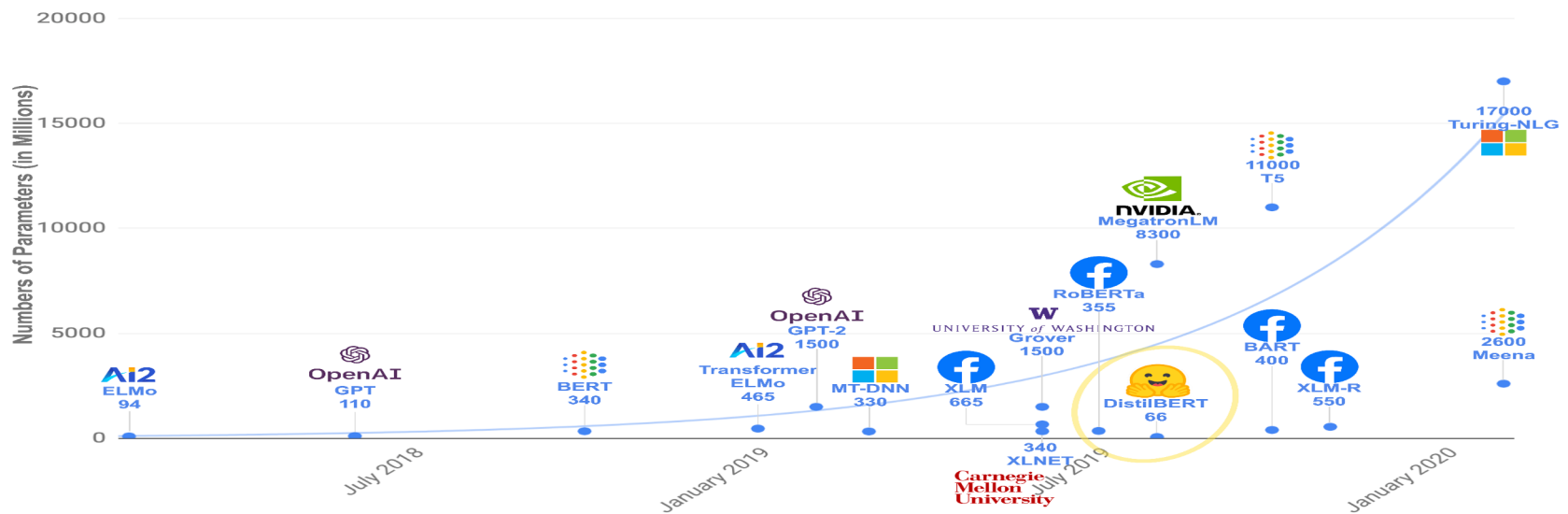


The [Transformer architecture](#) was introduced in June 2017. The focus of the original research was on translation tasks. This was followed by the introduction of several influential models, including:

- **June 2018:** [GPT](#), the first pretrained Transformer model, used for fine-tuning on various NLP tasks and obtained state-of-the-art results
- **October 2018:** [BERT](#), another large pretrained model, this one designed to produce better summaries of sentences (more on this in the next chapter!)
- **February 2019:** [GPT-2](#), an improved (and bigger) version of GPT that was not immediately publicly released due to ethical concerns
- **October 2019:** [DistilBERT](#), a distilled version of BERT that is 60% faster, 40% lighter in memory, and still retains 97% of BERT's performance
- **October 2019:** [BART](#) and [T5](#), two large pretrained models using the same architecture as the original Transformer model (the first to do so)
- **May 2020:** [GPT-3](#), an even bigger version of GPT-2 that is able to perform well on a variety of tasks without the need for fine-tuning (called *zero-shot learning*)

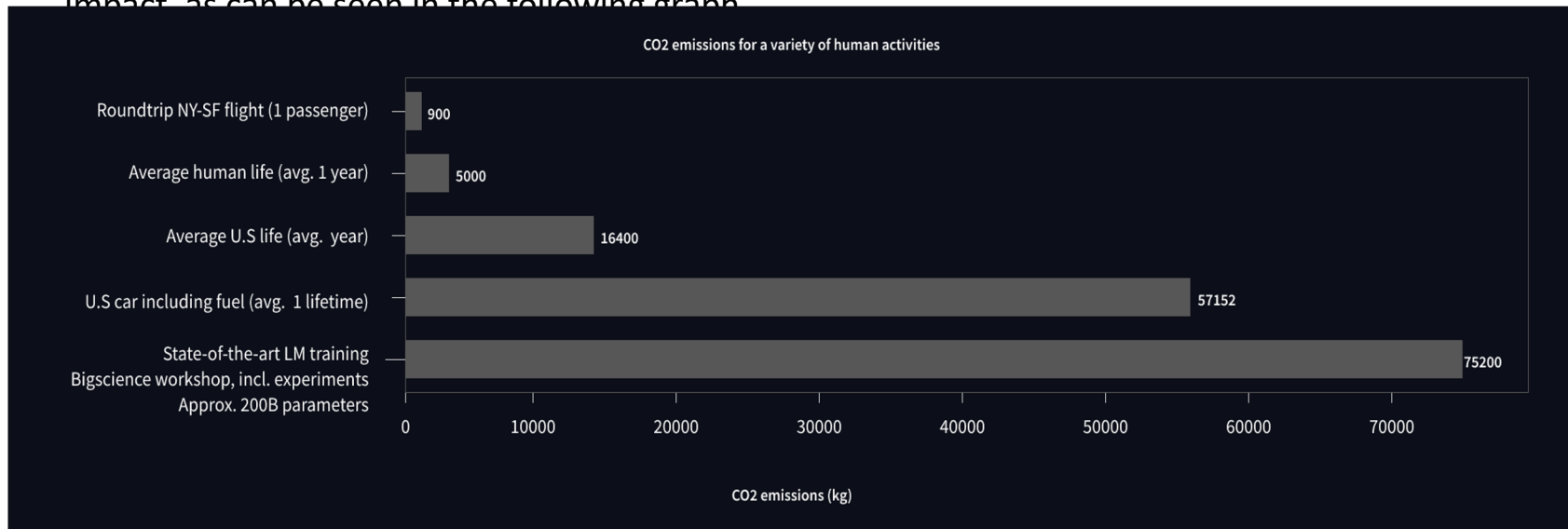
Transformer Models

Apart from a few outliers (like DistilBERT), the general strategy to achieve better performance is by increasing the models' sizes as well as the amount of data they are pretrained on.



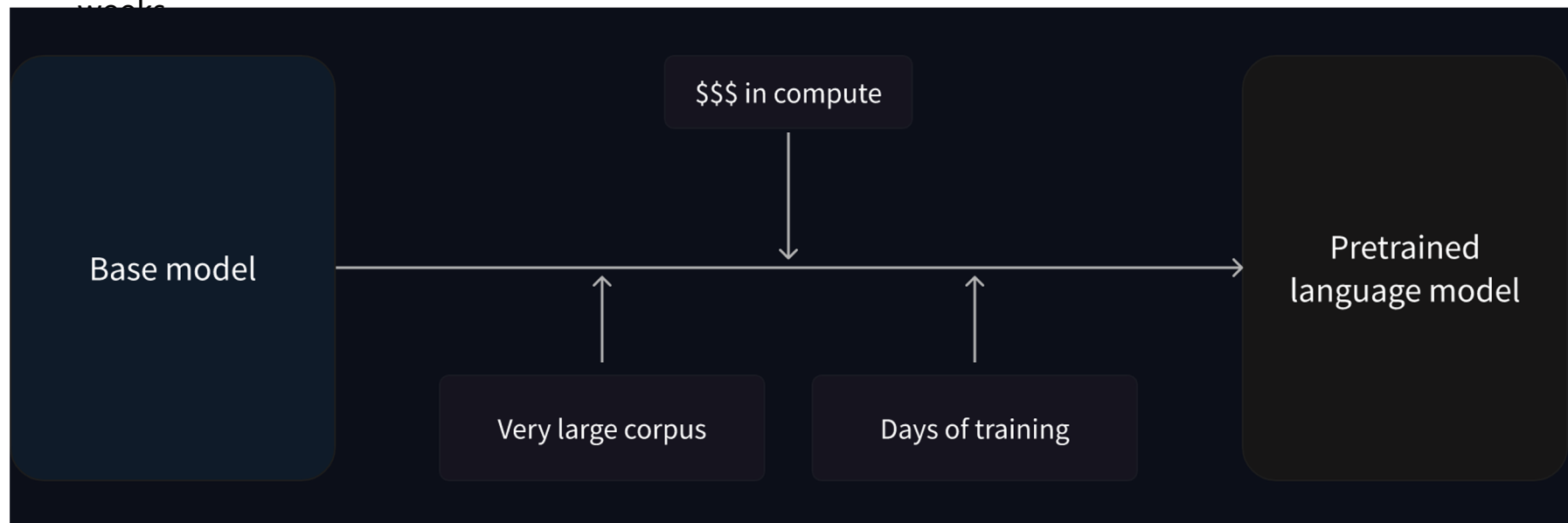
Transformers -Carbon Footprint

Unfortunately, training a model, especially a large one, requires a large amount of data. This becomes very costly in terms of time and compute resources. It even translates to environmental impact, as can be seen in the following graph



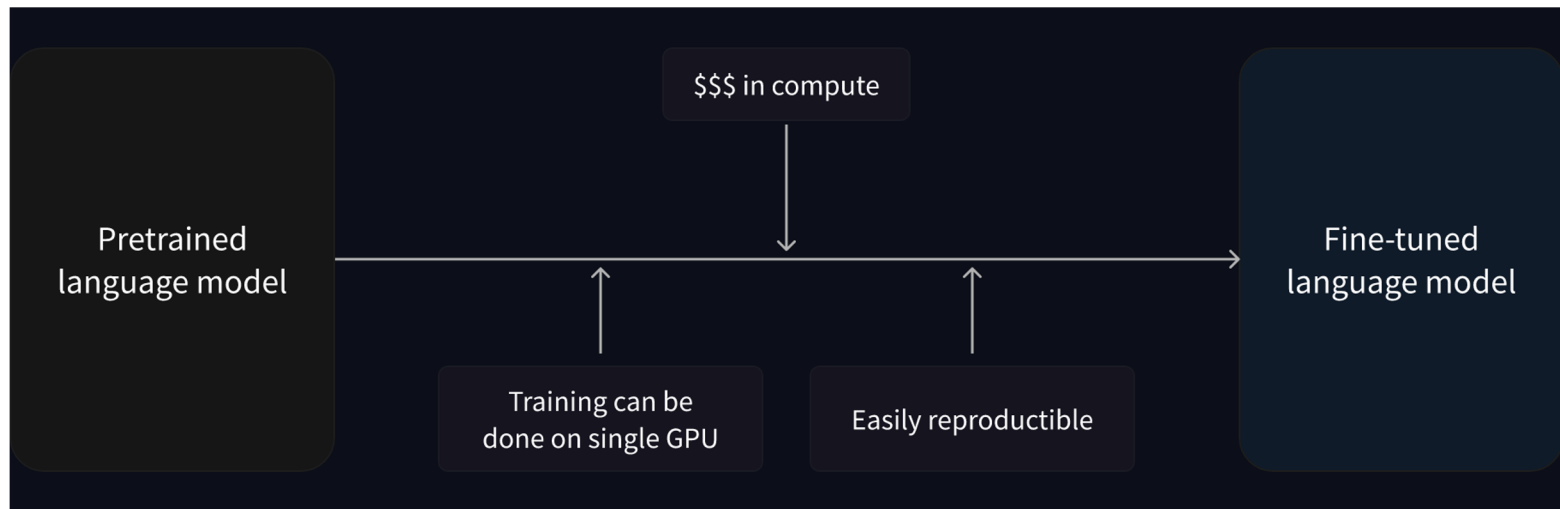
Transformers -Pretraining

Pretraining is the act of training a model from scratch: the weights are randomly initialized, and the training starts without any prior knowledge. This pretraining is usually done on very large amounts of data. Therefore, it requires a very large corpus of data, and training can take up to several weeks



Transformers -Finetuning

Fine-tuning, on the other hand, is the training done after a model has been pretrained. To perform fine-tuning, you first acquire a pre trained language model, then perform additional training with a dataset specific to your task.



Finetuning- why not simply train from scratch?

- The pretrained model was already trained on a dataset that has some similarities with the fine-tuning dataset. The fine-tuning process is thus able to take advantage of knowledge acquired by the initial model during pretraining (for instance, with NLP problems, the pretrained model will have some kind of statistical understanding of the language you are using for your task).
- Since the pretrained model was already trained on lots of data, the fine-tuning requires way less data to get decent results.
- For the same reason, the amount of time and resources needed to get good results are much lower.

Transformer Model Families

- GPT-like (also called *auto-regressive* Transformer models)
- BERT-like (also called *auto-encoding* Transformer models)
- BART/T5-like (also called *sequence-to-sequence* Transformer models)
- Encoder models use only the encoder of a Transformer model. At each stage, the attention layers can access all the words in the initial sentence. These models are often characterized as having “bi-directional” attention, and are often called auto-encoding models.
- Decoder models use only the decoder of a Transformer model. At each stage, for a given word the attention layers can only access the words positioned before it in the sentence. These models are often called *auto-regressive models*.
- Encoder-decoder models (also called sequence-to-sequence models) use both parts of the Transformer architecture. At each stage, the attention layers of the encoder can access all the words in the initial sentence, whereas the attention layers of the decoder can only access the words positioned before a given word in the input.